

DATABASE SYSTEM ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT

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Section:

Aggregation Queries (Q12 – Q20)

Query 12: Sort Customers by Balance (Descending)

Collection: customers

Stage 1 — \$sort:

```
{ "balance": -1 }
```

The screenshot shows the MongoDB Atlas interface for a project named 'Project 0'. The left sidebar displays the 'Data Explorer' with a tree view of clusters and collections. The main panel shows the 'Aggregations' tab for the 'Customers' collection. A single stage named '\$sort' is configured with the query '{ "balance": -1 }'. The 'Preview' button is active, showing a sample of 8 documents sorted by balance in descending order. The first two documents shown are:

```
{ "_id": ObjectId('6aad09660ce50839dd3da4cf'), "customer_id": 108, "name": "Meena Devi", "city": "Chennai", "account_type": "Savings", "balance": 320000, "status": "Active" }
```

```
{ "_id": ObjectId('6aad09660ce50839dd3da4cb'), "customer_id": 104, "name": "Sneha Rao", "city": "Warangal", "account_type": "Savings", "balance": 210000, "status": "Active" }
```

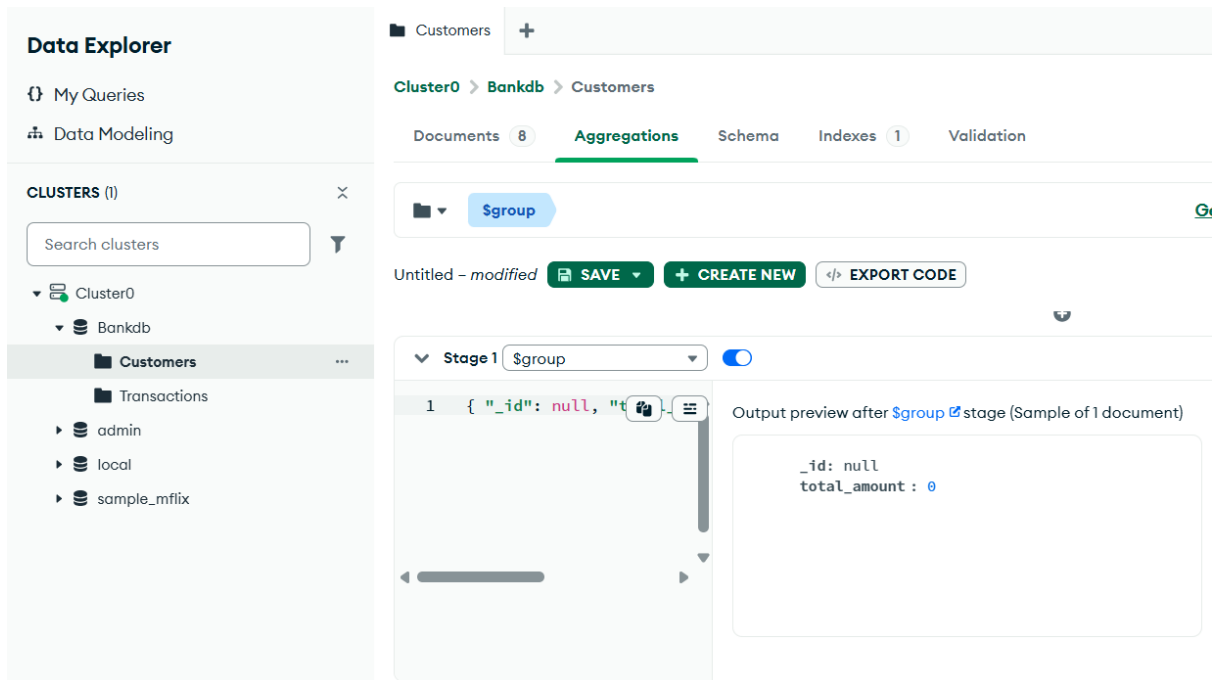
Query 13: Total Transaction Amount (All Transactions)

Collection:

transactions Stage 1

— \$group:

```
{ "_id": null, "total_amount": { "$sum": "$amount" } }
```



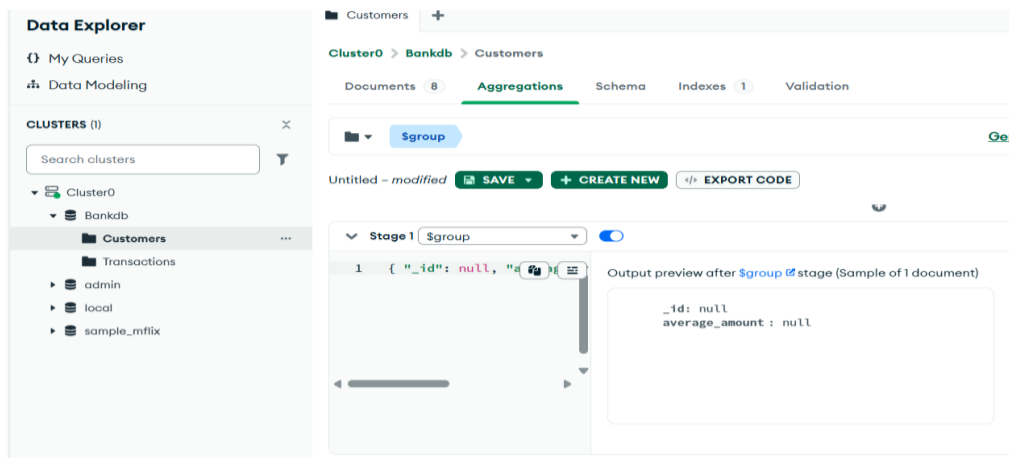
Query 14: Average Transaction Amount

Collection:

transactions Stage 1

— \$group:

```
{ "_id": null, "average_amount": { "$avg": "$amount" } }
```



Query 15: Total Amount Grouped by Transaction Type

Collection:

transactions Stage 1

— \$group:

```
{ "_id": "$type", "total_amount": { "$sum": "$amount" } }
```

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - Bankdb
 - Customers**
 - Transactions
 - admin
 - local
 - sample_mflix

Customers +

Cluster0 > Bankdb > Customers

Documents 8 Aggregations Schema Indexes 1 Validation

\$group

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Stage 1 \$group

1 { "_id": "\$type" }

Output preview after \$group stage (Sample of 1 document)

```
{
  "_id": null,
  "total_amount": 0
}
```

Query 16: Number of Transactions per Customer

Collection:

transactions Stage 1

— \$group:

```
{ "_id": "$customer_id", "transaction_count": { "$sum": 1 } }
```

Stage 2 — \$sort:

```
{ "transaction_count": -1 }
```

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - Bankdb
 - Customers**
 - Transactions
 - admin
 - local
 - sample_mflix

Customers +

Cluster0 > Bankdb > Customers

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\$group \$sort

Generate aggregation Explain Run Or

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Stage 1 \$group

1 { "_id": "\$customer_id" }

Output preview after \$group stage (Sample of 8 documents)

```
{
  "_id": 106,
  "transaction_count": 1
}
{
  "_id": 108,
  "transaction_count": 1
}
```

Stage 2 \$sort

1 { "transaction_count": -1 }

Output preview after \$sort stage (Sample of 8 documents)

```
{
  "_id": 104,
  "transaction_count": 1
}
{
  "_id": 106,
  "transaction_count": 1
}
```

Query 17: Join Transactions with Customer Details (\$lookup)

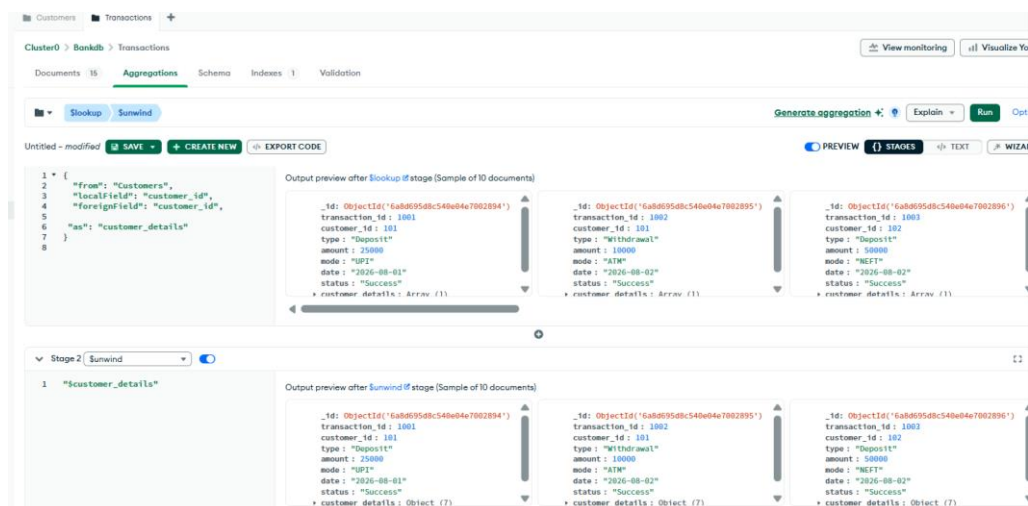
Collection: transactions Stage 1 — \$lookup:

```
{
  "from": "customers",
  "localField": "customer_id",
  "foreignField": "customer_id",

  "as": "customer_details"
}
```

Stage 2 — \$unwind:

```
"$customer_details"
```



Query 18: Display Customer Name And Transaction Amount

Collection:

transactions Stage 1

\$lookup:

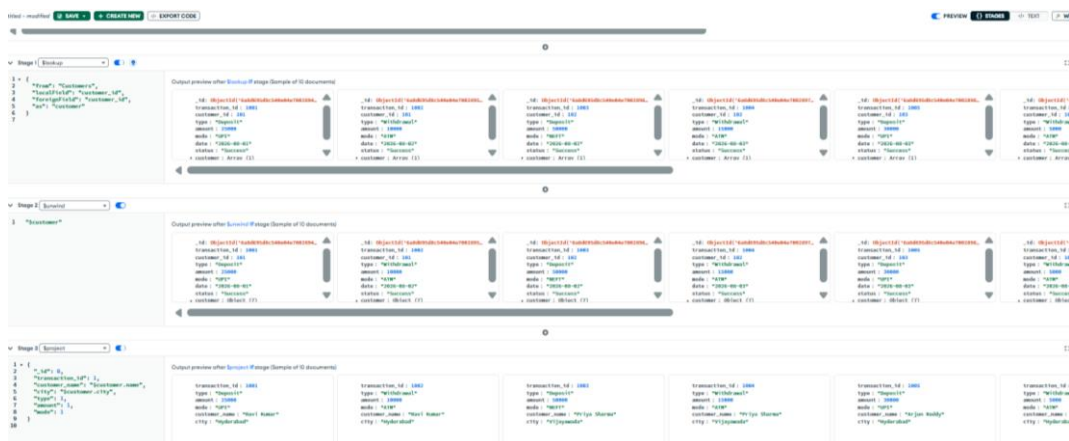
```
{
  "from": "customers",
  "localField": "customer_id",
  "foreignField": "customer_id",
  "as": "customer"
}
```

Stage 2 — \$unwind:

```
"$customer"
```

Stage 3 — \$project:

```
{
  "_id": 0,
  "transaction_id": 1,
  "customer_name": "$customer.name",
  "city": "$customer.city",
  "type": 1,
  "amount": 1,
  "mode": 1
}
```



Query 19: Total Transaction Amount For Each Customer

Collection:

transactions Stage 1

— \$group:

```
{ "_id": "$customer_id", "total_transaction": { "$sum": "$amount" } }
```

Stage 2 — \$lookup:

```
{
  "from": "customers",
  "localField": "_id",
  "foreignField": "customer_id",
  "as": "customer"
}
```

Stage 3 — \$unwind:

```
"$customer"
```

Stage 4 — \$project:

```
{
  "_id": 0,
  "customer_id": "$_id",
  "customer_name": "$customer.name",
  "total_transaction": 1
}
```

Stage 5 — \$sort:

```
{ "total_transaction": -1 }
```

The screenshot shows the Data Explorer interface with a pipeline configuration. The pipeline has four stages: \$project, \$lookup, \$unwind, and \$sort. The \$sort stage is selected, and its configuration is shown as follows:

```
1 { "total_transaction": -1 }
```

The output preview for the \$sort stage shows a sample of 10 documents, sorted by total_transaction in descending order. The documents are as follows:

transaction_id	type	amount	mode
1001	Deposit	25000	UPI
1012	Deposit	45000	IMPS
1003	Deposit	50000	NEFT
1010	Withdrawal	25000	ATM
1008	Withdrawal	25000	ATM
1006	Withdrawal	50000	ATM

Query 20: Complex Bank Transaction Report

Collection: customers

Stage 1 — \$match:

```
{ "status": "Active", "balance": { "$gt": 80000 } }
```

Stage 2 — \$lookup:

```
{
  "from": "transactions",
  "localField": "customer_id",
  "foreignField": "customer_id",
  "as": "transactions"
}
```

Stage 3 — \$unwind:

```
"$transactions"
```

Stage 4 — \$group:

```
{
  "_id": "$customer_id",
  "customer_name": { "$first": "$name" },
  "city": { "$first": "$city" },
  "balance": { "$first": "$balance" },
  "total_transactions": { "$sum": "$transactions.amount" },
  "transaction_count": { "$sum": 1 }
}
```

Stage 5 — \$sort:

```
{ "total_transactions": -1 }
```

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)
Search clusters

Cluster0
Bankdb
Customers

Transactions

Cluster0 > Bankdb > Customers

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Smatch Lookup Sumwind Sgroup Sort

Generate aggregation Explain Run Options

Preview STAGES TEXT WIZARD

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Stage1 Smatch

```
1 {
2   "status": "Active",
3   "balance": {
4     "igt": 8000
5   }
6 }
```

Output preview after Smatch if stage (Sample of 6 documents)

_id	customer_id	name	city	account_type	balance	status
ObjectId('6a8794ff45d5478318f8b94c')	181	Ravi Kumar	Hyderabad	Savings	8000	Active
ObjectId('6a8795ca5d5478318f8b94e')	182	Pritya Sharma	Vijayawada	Current	12000	Active
ObjectId('6a8795ca5d5478318f8b953')	187	Rahul Varma	Bangalore	Current	9500	Active
ObjectId('6a8795ca5d5478318f8b953')	187	Rahul Varma	Bangalore	Current	9500	Active
ObjectId('6a8795ca5d5478318f8b953')	187	Rahul Varma	Bangalore	Current	9500	Active
ObjectId('6a8795ca5d5478318f8b953')	187	Rahul Varma	Bangalore	Current	9500	Active

Stage2 Lookup

```
1 {
2   "from": "Transactions",
3   "localfield": "customer_id",
4   "foreignfield": "customer_id",
5   "as": "transactions"
6 }
7
```

Output preview after Lookup if stage (Sample of 6 documents)

_id	customer_id	name	city	account_type	balance	status	transactions
ObjectId('6a8794ff45d5478318f8b94c')	181	Ravi Kumar	Hyderabad	Savings	8000	Active	transactions: Array (2)
ObjectId('6a8795ca5d5478318f8b94e')	182	Pritya Sharma	Vijayawada	Current	12000	Active	transactions: Array (2)
ObjectId('6a8795ca5d5478318f8b953')	187	Rahul Varma	Bangalore	Current	9500	Active	transactions: Array (2)
ObjectId('6a8795ca5d5478318f8b953')	187	Rahul Varma	Bangalore	Current	9500	Active	transactions: Array (2)
ObjectId('6a8795ca5d5478318f8b953')	187	Rahul Varma	Bangalore	Current	9500	Active	transactions: Array (2)
ObjectId('6a8795ca5d5478318f8b953')	187	Rahul Varma	Bangalore	Current	9500	Active	transactions: Array (2)

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)
Search clusters

Cluster0
Bankdb
Customers

Transactions

Cluster0 > Bankdb > Customers

Documents 8 Aggregations Schema Indexes 1 Validation

Smatch Lookup Sumwind Sgroup Sort

Generate aggregation Explain Run Options

Preview STAGES TEXT WIZARD

Untitled - modify SAVE CREATE NEW EXPORT CODE

Stage3 Sumwind

```
1 "transactions"
```

Output preview after Sumwind if stage (Sample of 10 documents)

_id	customer_id	name	city	account_type	balance	status	transactions
ObjectId('6a8794ff45d5478318f8b94c')	181	Ravi Kumar	Hyderabad	Savings	8000	Active	transactions: Object (8)
ObjectId('6a8794ff45d5478318f8b94c')	181	Ravi Kumar	Hyderabad	Savings	8000	Active	transactions: Object (8)
ObjectId('6a8795ca5d5478318f8b94e')	182	Pritya Sharma	Vijayawada	Current	12000	Active	transactions: Object (8)
ObjectId('6a8795ca5d5478318f8b94e')	182	Pritya Sharma	Vijayawada	Current	12000	Active	transactions: Object (8)
ObjectId('6a8795ca5d5478318f8b953')	187	Rahul Varma	Bangalore	Current	9500	Active	transactions: Object (8)
ObjectId('6a8795ca5d5478318f8b953')	187	Rahul Varma	Bangalore	Current	9500	Active	transactions: Object (8)
ObjectId('6a8795ca5d5478318f8b953')	187	Rahul Varma	Bangalore	Current	9500	Active	transactions: Object (8)
ObjectId('6a8795ca5d5478318f8b953')	187	Rahul Varma	Bangalore	Current	9500	Active	transactions: Object (8)
ObjectId('6a8795ca5d5478318f8b953')	187	Rahul Varma	Bangalore	Current	9500	Active	transactions: Object (8)
ObjectId('6a8795ca5d5478318f8b953')	187	Rahul Varma	Bangalore	Current	9500	Active	transactions: Object (8)

Stage4 Sgroup

```
1 {
2   "$id": "$customer_id",
3   "$customer_name": {
4     "$first": "$name"
5   },
6   "$city": {
7     "$first": "$city"
8   },
9   "$balance": {
10    "$first": "$balance"
11  },
12  "$total_transactions": {
13    "$sum": "$transactions.amount"
14  },
15 }
```

Output preview after Sgroup if stage (Sample of 6 documents)

_id	customer_id	customer_name	city	balance	total_transactions	transaction_count
181	181	Sneha Rao	Murangal	12000	125000	2
181	181	Ravi Kumar	Hyderabad	8000	35000	2
181	181	Ravi Kumar	Hyderabad	8000	35000	2
181	181	Ravi Kumar	Hyderabad	8000	35000	2
181	181	Ravi Kumar	Hyderabad	8000	35000	2
181	181	Ravi Kumar	Hyderabad	8000	35000	2